Financial Document Intelligence Pipeline

System Architecture, Multimodal Ingestion, Extraction Engine & Independent Validation Layer

Architectural Mission: Automated enterprise financial workflows fail when heterogeneous documents (quarterly 10-Q statements, complex rental leases, loan covenants, commercial invoices) are flattened or directly routed to Large Language Models (LLMs). This incurs severe token waste, unvetted floating-point hallucinations, and arithmetic errors. The **Financial Document Intelligence Pipeline** resolves this by providing a multi-stage, production-grade engine: format-aware multimodal ingestion, hybrid semantic relevance classification (with all-MiniLM-L6-v2), decimal-safe financial extraction with scale preservation, and an **Independent Deterministic Accounting Validation & Quality-Gating Layer**.

1. End-to-End System Topology

The architecture is organized into seven decoupled, idempotent stages designed for high-throughput, auditable, and zero-data-loss execution:

Stage	Component	Key Responsibilities & Guarantees
Stage 1	**Ingestion & Security**	Magic-byte MIME verification, SHA-256 deduplication, zip-bomb ratio checks, filename sanitization, local raw storage.
Stage 2	**Multimodal Extraction**	PyMuPDF digital parsing, RapidOCR (ONNX Runtime) fallback for scanned pages, openpyxl for spreadsheets, python-docx for Word, layout coordinate preservation (bboxes).
Stage 3	**Structure-Aware Chunking**	Heading hierarchy preservation (H1/H2 breadcrumbs), table cell boundary preservation, paragraph grouping with stable hashing.
Stage 4	**Hybrid Semantic Relevance**	14-category financial ontology + hard negatives. Dense embedding (all-MiniLM-L6-v2) + FAISS CPU vector index + lexical signal heuristics. Three-way calibration: RELEVANT / UNCERTAIN / IRRELEVANT.
Stage 5	**Structured Financial Extraction**	Decimal-safe parsing (Decimal), currency & scale detection (USD Thousands -> 1,000x), row-level classification, parenthesized negative parsing, conditional fee preservation.
Stage 6	**Independent Quality Gate**	15 deterministic accounting & contractual rules. Evaluates Income Statement, Balance Sheet, signed change semantics, single-scale enforcement, and summary cross-checks. Decides ACCEPT / HOLD / REJECT.
Stage 7	**Handoff & React UI**	Structured JSON contract (/api/documents/{id}/report & /quality-gate), React dashboard with pipeline timeline, chunk explorer, and interactive accounting identity validation table.

2. Ingestion, Layout Preservation & Multimodal OCR

- **Dual-Path PDF Parsing:** Native digital PDFs are parsed using PyMuPDF (fitz) block inspection, preserving text spans, font metrics, line coordinates, and bounding boxes without floating-point artifacts. When native characters per page drop below 40, the page is automatically recognized as scanned and routed to RapidOCR (ONNX Runtime engine) with image deskewing.
- **Spreadsheets & Word Documents:** Preserves exact cell ranges (e.g. Sheet1!A1:F20), header rows, numeric formulas (retained strictly as untrusted text strings to prevent spreadsheet formula injection attacks), and hierarchical heading breadcrumbs.
- **Elimination of Numeric Table Corruption:** Custom cell boundary tokenization completely eliminates OCR comma-splitting bugs (e.g., previously malformed strings like $14200 , are cleanly extracted and preserved as $14,200).

3. Hybrid Semantic Relevance Engine

To avoid the dual pitfalls of keyword brittle matching and embedding over-generalization, FRIE employs a calibrated hybrid classification engine combining dense vector embeddings with lexical rule evidence:

Cat #	Category Slug	Financial Domain Scope
1	income_revenue	Gross/net revenues, software subscription revenues, consulting services, deferred revenue.
2	expenses_operating_costs	COGS, research & engineering, sales & marketing, G&A;, operational expenditures.
3	assets_liabilities_investments	Cash & equivalents, accounts receivable, PP&E; net, long-term debt, stockholders' equity.
4	financial_performance	Gross profit, operating income (EBIT), net income after tax, EBITDA, operating margins.
5	rent_leases_recurring_payments	Base monthly rent, security deposit terms, utility charges, lease escalations.
6	penalties_late_fees	Overdue interest, liquidated damages, conditional late fees ($75 or 5% unpaid balance).
7	contracts_payment_obligations	Legal covenants ('Tenant shall pay', 'reimbursement of expenses', consideration).
8	invoices_and_billing	Commercial invoices, billing line items, payment remittance details, Net 30/60 terms.
9	cash_flow_bank_transactions	Operating cash flows, deposits, wire transfers, bank statements, withdrawals.
10	non_financial (Hard Negatives)	Scientific charges, mineral deposits, code variables ($HOME, ${var}), NDAs, boilerplate text.

Embedding & Contrastive Similarity: Chunks are projected into 384-dimensional dense vectors using sentence-transformers/all-MiniLM-L6-v2. Cosine similarity is computed against indexed reference entries using FAISS. The final score aggregates top-2 positive category matches and penalizes hard-negative matches.

Document-Level Decisions: Aggregates chunk scores into four document classifications: FINANCIAL_RELEVANT (fully financial), MIXED_CONTENT (e.g. rental agreements with financial obligations), NEEDS_REVIEW (ambiguous or low-OCR pages), and NOT_FINANCIAL (cleanly filtered out without LLM costs).

4. Structured Financial Extraction & Decimal-Safe Normalization

- **Decimal-Safe Parsing:** Utilizes Python's exact decimal.Decimal library rather than binary floating-point arithmetic. Guarantees that values like $2,400.00 are parsed as 2400.00 and never corrupted into $240000USD.
- **Currency Scale Multipliers:** Detects document-level and table-level scale designations (e.g. Currency: USD (Thousands)). Correctly scales reported table values (14,200 $\to$ normalized amount 14,200,000.00) while preserving the exact verbatim string ($14,200), reported value (14200.00), and scale multiplier (1000.0) for auditable conversions.
- **Signed Accounting Values:** Converts parenthesized accounting numbers like ($8,370) into explicit signed decimals (-8370.00).
- **Change Column Semantics:** Automatically categorizes change columns as SIGNED_VALUE_DELTA (net change), MAGNITUDE_DELTA (expense growth like +$475 in COGS), PERCENTAGE_CHANGE (e.g. +11.1%), or BASIS_POINTS_DELTA (e.g. +110 bps), with explicit direction (INCREASE / DECREASE / FLAT).
- **Conditional Contract Clauses:** Accurately captures complex contractual clauses (e.g. $75.00 or 5% of the unpaid balance, whichever is greater) with structured condition metadata.

5. Independent Financial Validation & Quality-Gating Layer

The core innovation of the pipeline is the **Independent Financial Validation Engine** (app/validation.py). It evaluates extracted values completely independently from the extraction and OCR stages, ensuring no circular assumptions:

Core Principle of Validation Independence: Extraction confidence (OCR certainty) measures how well glyphs were recognized from page surfaces. Validation status measures whether the numbers make mathematical and semantic sense. A high extraction confidence (1.0) with broken balance sheet math is rejected; conversely, a low extraction confidence (0.65) with mathematically sound data passes validation.

Rule ID	Scope	Validation Formula / Logic	Policy Action
RULE_PARSE_INTEGRITY_001	Field	Validates digits are uncorrupted without orphan commas or detached punctuation.	FAILED / Review
RULE_CHANGE_SIGN_001	Field	Ensures parenthesized negative amounts are parsed with negative signs and direction.	Sets signed value
RULE_SCALE_UNIT_001	Field	Verifies scale multiplier applied exactly once; protects non-monetary metrics (%, bps) from scale.	FAILED on double-scale
RULE_CROSS_REF_001	Document	Cross-checks Executive Summary KPIs against Statement of Operations table rows across pages.	HOLD_FOR_REVIEW
RULE_ACCT_IS_REV_001	Income Stmt	Total Revenues == Software Subscription Revenue + Professional Services & Consulting.	Rounding Tol. check
RULE_ACCT_IS_GP_001	Income Stmt	Gross Profit == Total Revenues - Cost of Goods Sold (COGS).	Rounding Tol. check
RULE_ACCT_IS_OPEX_001	Income Stmt	Total Operating Expenses == Research & Engineering + Sales & Marketing + G&A.;	Rounding Tol. check
RULE_ACCT_IS_OPINC_001	Income Stmt	Operating Income (EBIT) == Gross Profit - Total Operating Expenses.	Rounding Tol. check
RULE_ACCT_IS_NETINC_001	Income Stmt	Net Income == Operating Income - Net Interest Expense - Income Tax Expense.	Rounding Tol. check
RULE_ACCT_BS_CA_001	Balance Sheet	Total Current Assets == Cash & Equivalents + Accounts Receivable Net + Prepaid Expenses.	Rounding Tol. check
RULE_ACCT_BS_TA_001	Balance Sheet	TOTAL ASSETS == Total Current Assets + PP&E; Net + Capitalized Software.	Rounding Tol. check
RULE_ACCT_BS_TL_001	Balance Sheet	Total Liabilities == Total Current Liabilities + Long-Term Borrowings.	Rounding Tol. check
RULE_ACCT_BS_EQUITY_001	Balance Sheet	TOTAL LIABILITIES & EQUITY == Total Liabilities + Stockholders' Equity & Retained Earnings.	Rounding Tol. check
RULE_ACCT_BS_EQ_001	Balance Sheet	Fundamental Accounting Equation: TOTAL ASSETS == TOTAL LIABILITIES & EQUITY.	Mismatch -> REJECT
RULE_RENTAL_TERMS_001	Rental Lease	Verifies presence of core contractual obligations: monthly rent, security deposit, late fee.	Contract verification

Quality-Gating Decisions:

- **ACCEPT:** All deterministic equations balance within tolerance ($0.01), scale is applied once, and no cross-reference conflicts exist.
- **ACCEPT_WITH_WARNINGS:** Non-critical minor subtotal items missing or warnings within acceptable threshold.
- **HOLD_FOR_REVIEW:** Cross-reference conflicts between executive summary KPIs and statement tables, or failed income statement components.
- **REJECT:** Fundamental balance sheet equation violation (Assets != Liabilities + Equity) or corrupted primary monetary values.

6. React UI Dashboard & Visual Quality Gate Inspector

The system includes a production-grade React (TypeScript) dashboard running on Vite with comprehensive operational and visual auditing interfaces:

View / Page	Route	Integrated Capabilities & Features
Quality Gate Inspector	/documents/:id	Live Quality Gate badge (ACCEPT/HOLD/REJECT), gate evaluation reasons, mathematical identities table showing expected vs actual values, and toggleable extracted field audit table.
Pipeline Timeline	/documents/:id	Visual stage progress (Queued -> Extracting -> Chunking -> Classifying -> Done) with job retry, reclassification triggers, and original file downloads.
Chunk Explorer	/documents/:id/chunks	Passage inspection with lexical signal highlights, bounding box overlays, table structure rendering, and human-in-the-loop reviewer feedback collection.
Reference Dataset	/reference	Curated vector database browser, category ontology distribution metrics, dynamic example creator, and FAISS index rebuild trigger.
Evaluation Suite	/evaluation	Real-time Precision-Recall curves, classification confusion matrix, OCR error diagnostics, and benchmark latency telemetry.

7. Public API Endpoints & Structured Contract Specifications

All data generated by the engine conforms to strictly typed Pydantic models with complete OpenAPI documentation:

HTTP Method & Route	Response Model	Contract Purpose & Payload Description
`GET /api/documents/{id}/quality-gate`	ValidationSummary	Returns quality gate decision, failed check counts, accounting validation calculations, cross-reference matches, and complete rule audit trail.
`GET /api/documents/{id}/structured-extraction`	StructuredExtractionReport	Full Section 7 report with chunk items, row-level classifications, normalized amounts, signed values, change semantics, and validation summary.
`GET /api/documents/{id}/relevant-chunks`	RelevantChunksResponse	Downstream extraction handoff payload containing relevant & uncertain chunks with untrusted_content: true flags and source provenance.
`GET /api/documents/{id}/report`	DocumentReport	Comprehensive document report combining document details, category breakdowns, chunk classifications, handoff, and structured extraction.

8. Test Suite Verification & Operational Metrics

The complete pipeline has been verified with a comprehensive 36-test automated regression suite covering all 18 validation scenarios, fixture regressions, and decimal-safe parsing:

Metric / Verification Target	Result / Operational Status
Automated Test Suite	**36 / 36 PASSED (100% Green)** across all regression & validation fixtures
Accounting Consistency Verification	22 deterministic checks passed on Financial Statements & Performance Report (Q3 2026/2025)
Rental Agreement Regression	Base rent ($2,400.00), security deposit ($2,400.00), and conditional late fee accurately captured
Embedding Model & Vector Index	sentence-transformers/all-MiniLM-L6-v2 (CPU-optimized, local FAISS index, 0 cloud dependencies)
Production Container Stack	Docker Compose orchestrating FastAPI backend (port 8000) and Vite/Nginx frontend (port 5173)